

UNIT III – SCIENTIFIC VALUES

1. What are scientific values?

Scientific values are basic principles that guide scientific thinking, research and communication. They help scientists work carefully, honestly and objectively.

2. What is scientific thinking?

Scientific thinking is the process of learning through observation, investigation and testing. It helps us solve problems and gain knowledge.

3. What is curiosity in scientific thinking?

Curiosity is the desire to ask questions about the world and search for answers.

4. What is observation?

Observation means carefully noticing and recording facts or events to understand a problem.

5. What is inductive reasoning?

Inductive reasoning moves from specific observations or examples to a general conclusion.

6. What is deductive reasoning?

Deductive reasoning starts with a general rule or statement and uses it to reach a specific conclusion.

7. Why is inductive reasoning useful?

It helps find patterns from observations and helps scientists form hypotheses and possible explanations.

8. Why is evidence important in science?

Evidence supports a claim or conclusion. It helps reduce personal bias and makes the conclusion more reliable.

9. Write a short note on hypotheses.

A hypothesis is a possible explanation or statement that can be tested using evidence or experiments.

10. How is a hypothesis tested?

A hypothesis is tested by collecting evidence, doing experiments or observations, analysing the results and checking whether the results support the hypothesis.

11. Mention any four types of inductive reasoning.

- Inductive generalization
 - Statistical induction
 - Causal reasoning
 - Sign reasoning
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12. Mention two applications of inductive and deductive reasoning.

- Science: scientists observe patterns, form hypotheses and test them.
- Everyday life: people use past experience to make assumptions and use known rules to make decisions.

13. Differentiate inductive and deductive reasoning.

Inductive reasoning	Deductive reasoning
Specific observations → general conclusion	General rule → specific conclusion
Conclusion is usually probable	Conclusion follows from the given premises when they are valid
Uses patterns from observations	Uses logic and known principles

14. What is skepticism?

Skepticism is an approach that encourages questioning and checking evidence before accepting a claim or conclusion.

15. What is empiricism?

Empiricism is the view that knowledge mainly comes from sensory experience, observation and evidence.

Easy flow: Observation → Question → Hypothesis → Testing → Evidence → Conclusion

UNIT IV – SOCIAL ETHICS

1. What is social ethics?

Social ethics is the study of moral issues in social structures, systems and communities.

2. What is ethical reasoning?

Ethical reasoning is a decision-making process used to judge what is right or wrong by considering benefits, harms and consequences.

3. How can ethical reasoning be applied to social problems?

Identify the problem, analyse its ethical issues, apply ethical principles, consider different values, make a decision, take action and evaluate the result.

4. Mention any four ethical principles used in social problems.

- Autonomy – respect a person's right to decide.
 - Beneficence – promote well-being.
 - Non-maleficence – avoid harm.
 - Justice – treat people fairly.
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5. Why is fairness important in social ethics?

Fairness helps distribute rights, opportunities and benefits equally and prevents unfair treatment.

6. What is gender bias?

Gender bias means treating people differently or unfairly because of their gender.

7. Mention any four issues related to gender bias.

- Unequal economic opportunities
 - Gender differences in education
 - Bias in healthcare
 - Under-representation in political leadership
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8. What is gender violence?

Gender violence is harmful or violent behaviour directed at a person because of gender. Examples include domestic violence and sexual violence.

9. Mention any four causes or practices linked with gender inequality and violence.

- Patriarchal traditions
 - Child marriage
 - Dowry system
 - Marginalization of women
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10. What is social discrimination?

Social discrimination means treating a person or group unfairly because of social identity or characteristics.

11. Mention the two major causes of social discrimination.

- Prejudice and stereotypes
 - Structural inequality
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12. What is constitutional protection?

Constitutional protection means rights and freedoms guaranteed by the Constitution, such as equality, liberty, privacy and freedom of religion.

13. Mention any four policies or measures that protect rights and reduce inequality.

- Legislation
 - Human rights commissions
 - Equality policies
 - Social welfare programmes
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14. What are inclusive practices?

Inclusive practices are strategies and actions that make people feel respected, valued and included, regardless of their background or characteristics.

15. Mention the five main inclusive practices.

- Understanding diversity
 - Creating an inclusive environment
 - Promoting equal opportunities
 - Challenging discrimination and bias
 - Education and training
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Easy process: Identify problem → Analyse → Apply ethics → Decide → Act → Evaluate

UNIT V – SCIENTIFIC ETHICS

1. What is scientific ethics?

Scientific ethics means the moral standards and practices that guide scientific research, reporting, interpretation and the use of scientific knowledge.

2. Why is scientific ethics important?

It protects research subjects, improves trust in science, supports reliable research and promotes responsible use of scientific knowledge.

3. Mention any four key aspects of scientific ethics.

- Integrity and honesty
 - Objectivity
 - Transparency
 - Respect for participants
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4. What is transparency in scientific pursuits?

Transparency means clearly sharing research methods, procedures, data and relevant information so that others can understand and verify the work.

5. What is fairness in scientific pursuits?

Fairness means giving equal opportunities and treating researchers and participants with dignity, respect and justice.

6. Mention four practices that improve transparency.

- Open communication
 - Data sharing
 - Disclosure of conflicts of interest
 - Public communication of findings
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7. Mention four practices that promote fairness.

- Equal opportunities
 - Inclusive practices
 - Ethical treatment of participants
 - Fair peer review
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8. Differentiate transparency and fairness in scientific pursuits.

Transparency	Fairness
Focuses on openness and clarity in research	Focuses on equal and respectful treatment

Includes methods, data and conflict disclosure	Includes equal opportunities and ethical treatment
Helps others understand and verify research	Helps prevent discrimination and unfair treatment

9. What are unfair practices in scientific research?

Unfair practices are actions that make research dishonest or unreliable, such as biased reporting, manipulation of data or copying another person's work.

10. Why should researchers avoid bias?

Bias can affect the way information is collected, analysed and interpreted. Avoiding bias improves objectivity and reliability.

11. Mention any four scientific inventions or developments that benefit society.

- Medicine and healthcare
 - Communication and information technology
 - Renewable energy and sustainability
 - Agricultural advancements
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12. How do scientific inventions improve society?

They improve healthcare, communication, education, agriculture, energy, transportation and the quality of life.

13. Mention three scientific developments that help us understand the world.

- Space exploration
 - Scientific research
 - Scientific instruments
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14. Write a short note on the responsible use of scientific knowledge.

Scientists should use scientific knowledge and technology for the benefit of society and avoid uses that may cause harm.

15. Mention any four ethical issues in science and technology.

- Unfair or biased use of technology
 - Environmental degradation
 - Erosion of public trust
 - Health or social inequality caused by technology
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Easy idea: Ethical research → Reliable knowledge → Public trust → Benefit to society